

Week 8 Workshop

Question 1

Let F_{t-1} denote the information set available up to time $t - 1$. Define the following concepts:

- (i) A white noise process.
- (ii) An MDS

Briefly explain the difference between white noise and a martingale difference sequence.

Question 2

Suppose that u_t is an MDS.

- (i) Show that the unconditional mean of u_t is zero.
- (ii) Show that u_t has no autocorrelation.

Question 3

The data (*Week 8 q3.xlsx*) were generated from an unknown autoregressive process. Estimate AR(p) models for $p = 1, \dots, 5$, compare the AIC values, and select the most suitable model. After selecting the model using AIC, check whether the residuals look like white noise using the correlogram.